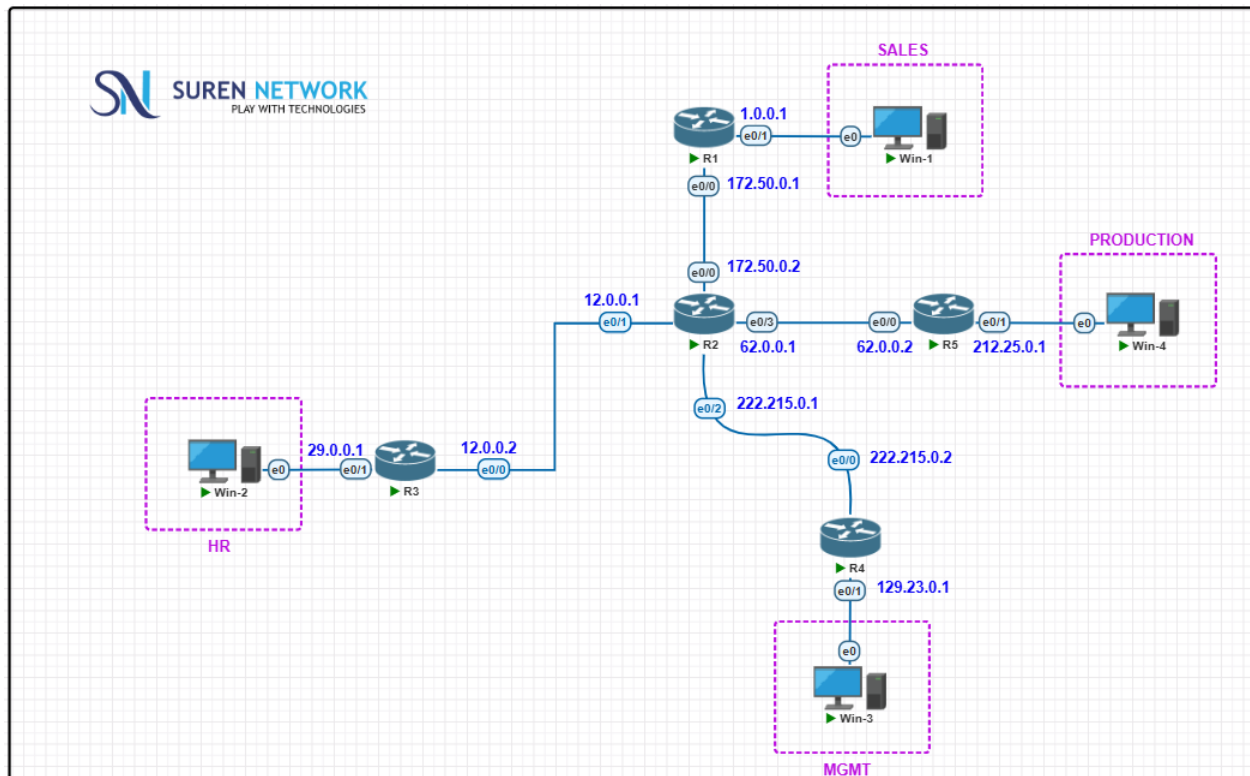




Assign Ip address in all cisco routers

<pre> R3 do sh ip int br Interface IP-Address OK? Method Status Ethernet0/0 12.0.0.2 YES TFTP up Ethernet0/1 29.0.0.1 YES TFTP up Ethernet0/2 unassigned YES TFTP down Ethernet0/3 unassigned YES TFTP down </pre>	<pre> R4 do sh ip int br Interface IP-Address OK? Method Status Ethernet0/0 222.215.0.2 YES TFTP up Ethernet0/1 129.23.0.1 YES TFTP up Ethernet0/2 unassigned YES TFTP down Ethernet0/3 unassigned YES TFTP down </pre>
<pre> R1 do sh ip int br Interface IP-Address OK? Method Status Ethernet0/0 172.50.0.1 YES manual up Ethernet0/1 1.0.0.1 YES TFTP up Ethernet0/2 unassigned YES TFTP down Ethernet0/3 unassigned YES TFTP down R1(config-if)# </pre>	<pre> R2 do sh ip int br Interface IP-Address OK? Method Status Ethernet0/0 172.50.0.2 YES TFTP up Ethernet0/1 12.0.0.1 YES TFTP up Ethernet0/2 222.215.0.1 YES TFTP up Ethernet0/3 62.0.0.1 YES TFTP up R2(config)# </pre>
<pre> R5 do sh ip int br Interface IP-Address OK? Method Status Ethernet0/0 62.0.0.2 YES TFTP up Ethernet0/1 212.25.0.1 YES TFTP up Ethernet0/2 unassigned YES TFTP down Ethernet0/3 unassigned YES TFTP down </pre>	

Add static route in all cisco routers

```
R3
S 1.0.0.0/8 [1/0] via 12.0.0.1
C 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 12.0.0.0/8 is directly connected, Ethernet0/0
L 12.0.0.2/32 is directly connected, Ethernet0/0
C 29.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 29.0.0.0/8 is directly connected, Ethernet0/1
L 29.0.0.1/32 is directly connected, Ethernet0/1
S 62.0.0.0/8 [1/0] via 12.0.0.1
S 129.23.0.0/16 [1/0] via 12.0.0.1
S 176.50.0.0/16 [1/0] via 12.0.0.1
C 212.25.0.0/16 [1/0] via 12.0.0.1

R1
C 1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 1.0.0.0/8 is directly connected, Ethernet0/1
L 1.0.0.1/32 is directly connected, Ethernet0/1
S 12.0.0.0/8 [1/0] via 172.50.0.2
S 29.0.0.0/8 [1/0] via 172.50.0.2
S 62.0.0.0/8 [1/0] via 172.50.0.2
C 129.23.0.0/24 is subnetted, 1 subnets
L 129.23.0.0 [1/0] via 172.50.0.2
C 172.0.0.0/8 is directly connected, Ethernet0/0
L 172.50.0.0/32 is subnetted, 1 subnets
L 172.50.0.1 is directly connected, Ethernet0/0
S 212.25.0.0/24 [1/0] via 172.50.0.2
S 222.215.0.0/24 [1/0] via 172.50.0.2
R1(config-if)#

R5
S 1.0.0.0/8 [1/0] via 62.0.0.1
S 12.0.0.0/8 [1/0] via 62.0.0.1
S 22.0.0.0/24 is subnetted, 1 subnets
L 22.0.0.0 [1/0] via 62.0.0.1
S 29.0.0.0/8 [1/0] via 62.0.0.1
C 62.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 62.0.0.0/8 is directly connected, Ethernet0/0
L 62.0.0.2/32 is directly connected, Ethernet0/0
S 129.23.0.0/16 [1/0] via 62.0.0.1
S 172.50.0.0/16 [1/0] via 62.0.0.1
S 212.25.0.0/24 is variably subnetted, 2 subnets, 2 masks
L 212.25.0.0/24 is directly connected, Ethernet0/1
L 212.25.0.1/32 is directly connected, Ethernet0/1
R5(config-if)#

R4
Gateway of last resort is not set
S 1.0.0.0/8 [1/0] via 222.215.0.1
S 12.0.0.0/8 [1/0] via 222.215.0.1
S 29.0.0.0/8 [1/0] via 222.215.0.1
S 62.0.0.0/8 [1/0] via 222.215.0.1
C 129.23.0.0/16 is variably subnetted, 2 subnets, 2 masks
L 129.23.0.0/16 is directly connected, Ethernet0/1
L 129.23.0.1/32 is directly connected, Ethernet0/1
S 172.50.0.0/16 [1/0] via 222.215.0.1
S 212.25.0.0/24 [1/0] via 222.215.0.1
C 222.215.0.0/24 is variably subnetted, 2 subnets, 2 masks
L 222.215.0.0/24 is directly connected, Ethernet0/0
L 222.215.0.2/32 is directly connected, Ethernet0/0
R4(config-if)#

R2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set
S 1.0.0.0/8 [1/0] via 172.50.0.1
C 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 12.0.0.0/8 is directly connected, Ethernet0/1
L 12.0.0.1/32 is directly connected, Ethernet0/1
S 29.0.0.0/8 [1/0] via 12.0.0.2
C 62.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 62.0.0.0/8 is directly connected, Ethernet0/3
L 62.0.0.1/32 is directly connected, Ethernet0/3
S 129.23.0.0/16 [1/0] via 222.215.0.2
C 172.50.0.0/16 is variably subnetted, 2 subnets, 2 masks
L 172.50.0.0/16 is directly connected, Ethernet0/0
L 172.50.0.2/32 is directly connected, Ethernet0/0
S 212.25.0.0/24 [1/0] via 62.0.0.2
C 222.215.0.0/24 is variably subnetted, 2 subnets, 2 masks
L 222.215.0.0/24 is directly connected, Ethernet0/2
L 222.215.0.1/32 is directly connected, Ethernet0/2
R2(config)#
```

Enable dhcp server for All department On R2

```
R2
R2(config)#
R2(config)#ip dhcp pool SALES
R2(dhcp-config)#network 1.0.0.0 255.0.0.0
R2(dhcp-config)#default-router 1.0.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#lease 1 2 3
R2(dhcp-config)#ex
R2(config)#ip ex
R2(config)#ip dhcp pool PRODUCTION
R2(dhcp-config)#network 212.25.0.0 255.255.255.0
R2(dhcp-config)#default-router 212.25.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#lease 1 2 3
R2(dhcp-config)#ex
R2(config)#
R2(config)#ip dhcp pool MGMT
R2(dhcp-config)#network 129.23.0.0 255.255.0.0
R2(dhcp-config)#default-router 129.23.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#lease 1 2 3
R2(dhcp-config)#ex
R2(config)#
R2(config)#ip dhcp pool HR
R2(dhcp-config)#network 29.0.0.0 255.0.0.0
R2(dhcp-config)#default-router 29.0.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#lease 1 2 3
R2(dhcp-config)#ex
R2(config)#
R2(config)#do wr
Building configuration...
[OK]
R2(config)#
```

Enable dhcp relay agent on All router

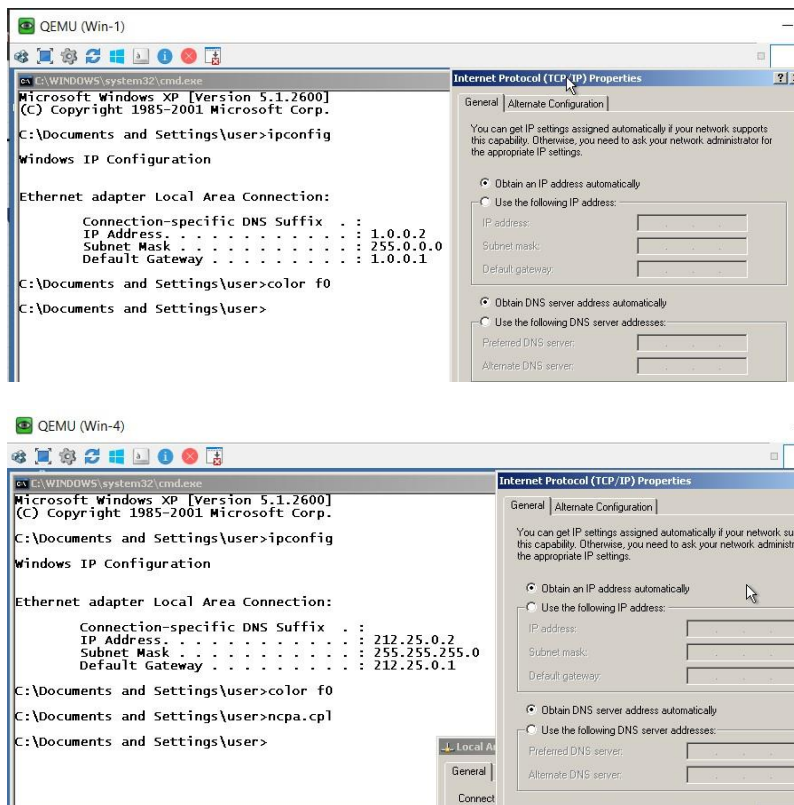
```
R1
R1(config)#int e0/1
R1(config-if)#ip hel
R1(config-if)#ip help
R1(config-if)#ip helper-address 172.50.0.2
R1(config-if)#
R1(config-if)#

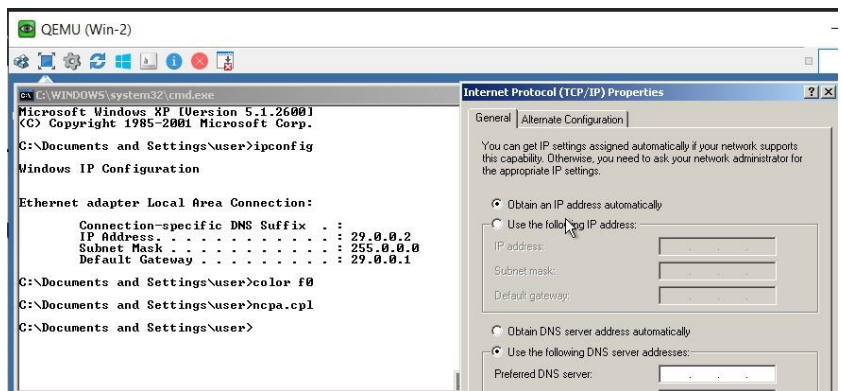
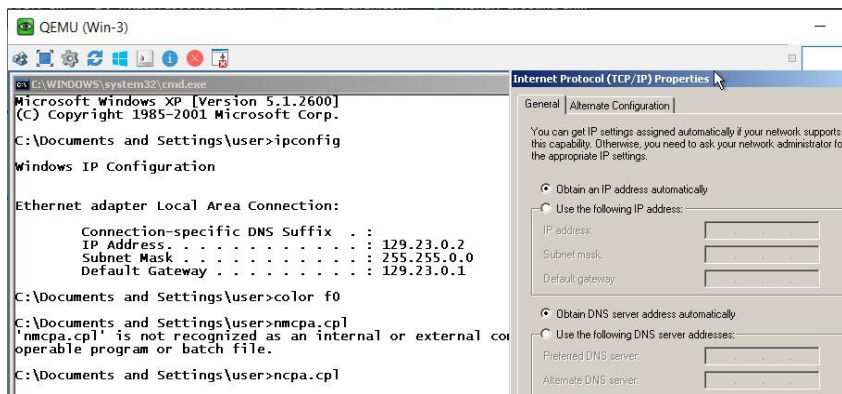
R5
R5(config)#int e0/1
R5(config-if)#ip helper-address 62.0.0.1
R5(config-if)#
R5(config-if)#do wr
Building configuration...

R4
R4(config)#
R4(config)#int e0/1
R4(config-if)#ip helper-address 222.215.0.1
R4(config-if)#

R3
R3(config)#int e0/1
R3(config-if)#ip helper-address 12.0.0.1
R3(config-if)#
R3(config-if)#do wr
Building configuration...
[OK]
```

Get Ip from the DHCP server (R2)





Check dhcp database in Server (R2)

